

risk/reward performing arts festival 2025

Autumn Jing and Clutch Anderson

Summary: An audiovisual interactive installation that explores the interaction of light, shaped by concepts of physical phenomena, spectral data, and scientific principle, all within a performative context.

AS LIGHT FILTERS IN, an audiovisual installation demonstrates interactions between photons, light-responsive forms, electronics, and sound.

A field of prisms, light paths, soundscape, interactivity, illumination, total internal reflection. Grazing incidence and emergence.

We will create an interactive installation that explores the interaction of light, shaped by concepts of physical phenomena, spectral data, and scientific principle, all within a performative context. Electronic components for the installation will be derived from Autumn's current project *uuutopiaaa*, as part of PICA's precipice fund pool. While both projects harvest the same material, the differences are recognizable in form, function, and communicative design, with the technical aspects maintaining a consistent underlying functionality.

Studies of light reveal connections to sound synthesis as a whole. These connections exist both as scientific fact as well as playful artistic inferences. To artists, the explorative similarities between light and sound are abundant. Regarding performance and show, light and sound are foundational to all forms of sensory activation. Our goal is to investigate scientific principles and derive performative artistic explorations from them.

In our installation, light will move and separate. Light will collide and diffract through a variety of uniquely shaped objects. Light will infiltrate sensors. Light will trigger and manipulate audio samples, and will be responsible for controlling audio effects. This play between light and audio will be the creative result of investigating the similarities between light and sound. We will lead the audience through demonstrations that build upon concepts of diffraction, harmonics, interference, and resonance. The ways in which light and sound relate to these concepts will be made transparent to the audience.

One main principle we are investigating is the process of diffraction and diffraction grating. Diffraction is the process in which light bends around an obstacle in its path. In optics, materials known as diffraction gratings are used to separate and sort waves, thus amplifying specific wavelengths of white light. Diffraction gratings are manufactured to feature periodic edges (sometimes resembling a sawtooth pattern) that correspond to specific wavelengths of light waves. Instead of light beams *reflecting* off of the surface, the jagged surface profile allows specific wavelengths of light to diffract at different angles. The distinct diffraction grating

profiles determine the resulting angles of similar-wavelength waves, called “orders”. In summary, different profile patterns determine the resulting angles (orders) of diffracted light.

This matters to us because it guides us to corresponding audio processing techniques. We see how the spectrum of light changes due to different physical structures. When considering audio effect theory, we will mirror this concept, using things like convolution reverbs and resonators to mimic a diffractive surface.

All of the audio treatment processes will be in response to some conceptual knowledge explored in the previous section. Recall, for example, the jagged surfaces used to manufacture diffraction gratings. They happen to be modeled after the basic foundational shapes of synthesis, resembling sine, sawtooth, and square waves. This will in turn serve as our foundation for the demonstration and performance. Recall again how using a different grating profile (I.E, a different waveform) affects the resulting angles, or orders, of diffracted light. This concept also holds true for audio synthesis. Sine waves contain one harmonic, square waves contain only odd harmonics, and sawtooth waves contain both odd and even harmonics. If we consider our basic waveforms as physical profiles, and remain transparent to the audience in this consideration, we will be emphasising the physical relationship between light and sound at each stage of the performance. Treating our pre-loaded .wav files with physical modeling effects will be key in demonstrating the concept of diffraction. By using these effects, the audience will be aware of the concepts and inspirations we are working from. Enabling that transparency is not only our top priority, it is our responsibility as performers!

For the Risk/Reward Festival, Autumn will develop a prototype of a digital waveform synthesizer using a designed system of ESP32 microcontrollers. This iteration will function as a sampler that loads .wav files, applying real-time audio manipulation effects to explore sound design and creative audio processing. This is a condensed iteration of a larger ongoing project, which ultimately focuses on generating pure waveforms via hardware processing with proximity sensor input for modulation.

The system will allow for various audio effects, such as pitch shifting, time-stretching, and filtering, simulating a high-level version of the sound manipulation envisioned in the final iteration of the larger project. Instead of using proximity sensors for interaction, this project will rely on light as the input medium, offering an innovative way to control and modulate the audio output in real-time.

The focus of this project is to showcase the potential of the system in both functionality and interactivity, which aligns with the larger vision of hardware-based interaction. Revealing the connections between light and sound design will expand possibilities for performance, experimentation, and expression.

The audience will be drawn into a speculative experience of light and sound in high contrast. As light is shaped by varying structures and geometric formations, it sparks curiosity: *What processes are unfolding? Why does the light transform? How will these connections influence my creativity?* This interplay between light and sound invites deeper inquiry, encouraging the audience to question the physical mechanisms behind the shifting forms and investigate the unexpected connections between the two mediums.

Our R&D will continue our deep-dive into the numerous primordial connections between light and sound, which will then be presented through live demonstrations of the technical processes at work. We aim to create an interactive space where experimentation unfolds in real time, fostering a collaborative exchange of knowledge between us as presenters and the audience as active participants. By sharing our findings with transparency, we invite viewers to engage, question, and explore the evolving relationship between these two dynamic mediums.

BIOGRAPHIES

<http://signalkitten.net/> + <https://www.linkedin.com/in/autumngreenbean/>

AUTUMN BIO

Autumn Jing is an audiovisual designer who specializes in signal processing and interactive media. With a background in photography, videography, and sound design, she combines analog and digital techniques to produce experimental works. Awarded the 2024 PICA Precipice Fund grant, Autumn is leading the development of a proximity-sensitive, hardware-based sound synthesizer in collaboration with a 3D sculpture installation. Autumn received her BA in Art and Technology from the University of Oregon in 2023. Her work has been shown at the JSMA, the Eugene Riverfront, and the Eugene opera. Autumn works under the alias *signalkitten*, where she creates work that integrates multi-layered signal-processing, such as analog VHS feedback, audiovisual hardware development, and waveform sound synthesis.

CLUTCH BIO

<https://vimeo.com/clutch>

Clutch Anderson is a Portland-based artist working in audio, video, and animation. His work is an experimental process driven by post-processing and resampling. Relying on dithering, granular synthesis, and digital artifacts, Clutch creates audiovisual works which investigate dreamlike spaces. These investigations include ambient/experimental/glitch albums, multi-channel video installations, and research writings. Clutch shares these projects under the artist name clutch YUP.